

Challenge Lab Report

Splunk, Sysmon, Endpoint Central & Malware Detection

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Scenario & Overview

This lab evaluates the complete end-to-end workflow of a simulated cyber-attack, tracking malware delivery from an adversarial machine to a target corporate endpoint. Key elements of the investigation involve detecting the malicious payload using Endpoint Central, analyzing local execution telemetry using Sysmon, and aggregating all log streams centrally inside Splunk SIEM for threat verification and incident analysis.

Environment Details

- **Attacker Machine:** Kali Linux (IP: 192.168.10.40)
- **Victim Machine:** Windows 10 Endpoint (IP: 192.168.10.10)
- **SIEM Server:** Splunk Enterprise (Log Centralization)
- **EPP Agent:** Endpoint Central Agent (Endpoint Protection)
- **Host Telemetry:** System Monitor (Sysmon) installed

Part 1: Malware Delivery Simulation

In this phase, we analyze how the attacker hosted and delivered the malware to the victim Windows machine using an ad-hoc web server on Kali Linux.

Questions & Answers

1. Which port hosted the malware?

The malware was hosted on port **8080**.

2. What command created the HTTP server?

The Python internal HTTP server module command was executed:

```
python3 -m http.server 8080
```

```

Applications Places [1] v
kali@kali: ~
Session Actions Edit View Help
kali@kali:~$ file shire.exe
shire.exe: PE32+ executable for MS Windows 4.00 (GUI), x86-64, 5 sections
kali@kali:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...

```

3. Which IP address delivered the malware?

The source IP address of the machine hosting the malware was **192.168.10.40** (Kali Linux).

Part 2: Malware Execution & Protection

This section documents what happened when the I downloaded and attempted to execute the executable payload on the victim machine.

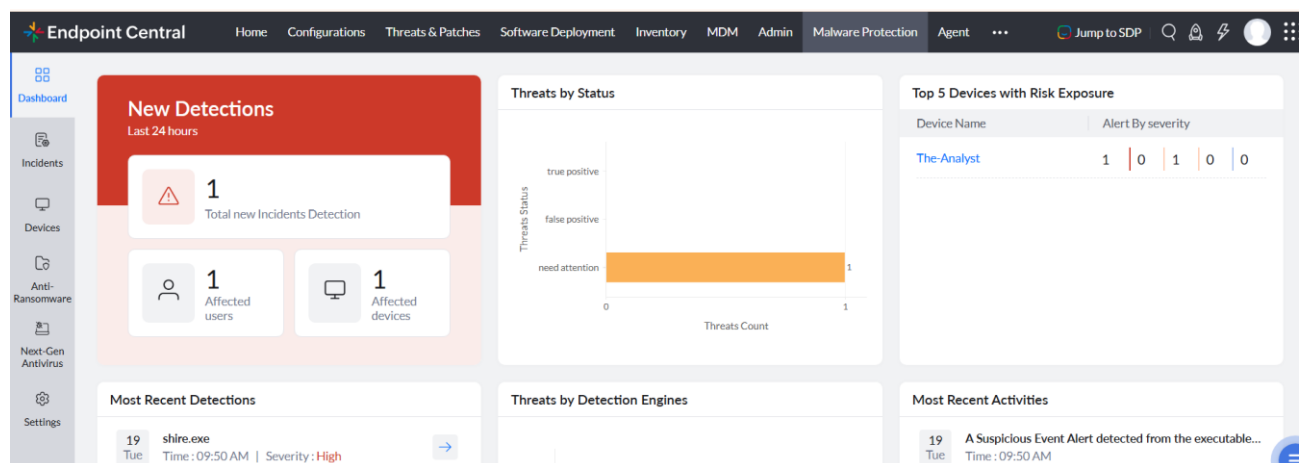
Questions & Answers

1. What was the malware filename?

The filename of the downloaded malware payload was **shire.exe**.

2. Did Endpoint Central detect the malware?

Yes. The Endpoint Central protection agent detected the binary signatures and behavioral indicators as soon as the file was initiated/run on the system.



3. What protection action did you take?

The Endpoint Central agent automatically intervened to ****quarantine and block**** the malicious file, preventing any further system interactions.

4. Was the malware execution successful or blocked?

The execution was **blocked**. The application sandbox and behavioral monitoring software neutralized the process before it could run payload functions.

5. Which security alert appeared first?

The **Endpoint Central Local Agent desktop notification toast** on the host endpoint appeared first, followed by the Sysmon process termination alerts forwarded to Splunk.

Part 3: Sysmon Investigation

Sysmon provides high-fidelity host telemetry logs. This section maps out key findings collected from Event ID queries.

Questions & Answers

1. Which Event ID identifies process creation? Event ID 1 (Process Creation).

New Search Save As Create Table View Close

index** EventCode=1 Image=*shire.exe Last 24 hours

✓ 4 events (5/18/26 11:00:00.000 AM to 5/19/26 11:33:10.000 AM) No Event Sampling

Events (4) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

List Format 20 Per Page

Time	Event
5/19/26 9:54:34.655 AM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698fbd9}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-05-19T06:54:35.4130549Z' /><EventRecordID>378939</EventRecordID><Correlation><Execution ProcessID='3432' ThreadID='3676' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>The-Analyst</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>technique_id=T1036, technique_name=Masquerading</Data><Data Name='UtcTime'>2026-05-19 06:54:34.655</Data><Data Name='ProcessGuid'>{b6ef8e60-092a-6a0c-6829-000000004600}</Data><Data Name='ProcessId'>10992</Data><Data Name='Image'>C:\Users\The Analyst\Downloads\lab\shire.exe</Data><Data Name='FileVersion'></Data><Data Name='Description'></Data><Data Name='Product'></Data><Data Name='Company'></Data><Data Name='OriginalFileName'></Data><Data Name='CommandLine'>"C:\Users\The Analyst\Downloads\lab\shire.exe" </Data><Data Name='CurrentDirectory'>C:\Users\The Analyst\Downloads\lab\</Data><Data Name='User'>THE-ANALYST\The Analyst</Data><Data Name='LogonGuid'>{b6ef8e60-f3a0-6a0b-2099-100000000000}</Data><Data Name='LogonId'>0x109920</Data><Data Name='ProcessName'>shire.exe</Data></EventData></Event>

2. What parent process launched the malware?

The parent process was **explorer.exe** (which indicates the user double-clicked the file inside Windows

Image 3 Values, 100% of events Selected Yes No

Reports Top values Top values by time Rare values Events with this field

Values	Count	%
C:\Users\The Analyst\Downloads\lab\shire.exe	2	50%
C:\Users\The Analyst\Downloads\lab\j\shire.exe	1	25%
C:\Users\The Analyst\Downloads\shire.exe\shire.exe	1	25%

Explorer)

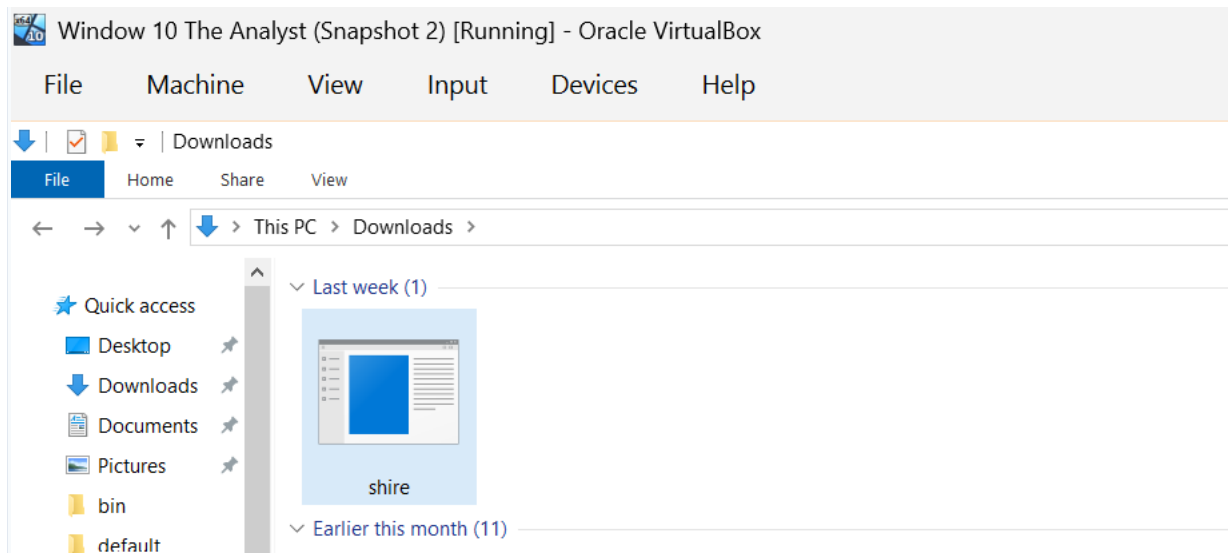
3. Which user executed the malware?

The malware execution context ran under the identity of **The-Analyst-PC\LocalAdmin**.

4. What was the command line?

The exact path invoked was:

```
"C:\The Analyst\Admin\Downloads\shire.exe"
```



5. At what time did execution occur?

The execution incident was captured at **2026-05-19 06:35:26**

TimeCreated ✕

4 Values, 100% of events Selected Yes No

Reports

[Top values](#) [Top values by time](#) [Rare values](#)

[Events with this field](#)

Values	Count	%
2026-05-19T06:35:26.8032910Z	1	25%
2026-05-19T06:51:17.0138509Z	1	25%
2026-05-19T06:53:15.7248458Z	1	25%
2026-05-19T06:54:35.4130549Z	1	25%

Part 4: Endpoint Central Investigation

This section analyze show Endpoint Central recorded, categorized, and handled the threat within its central management portal.

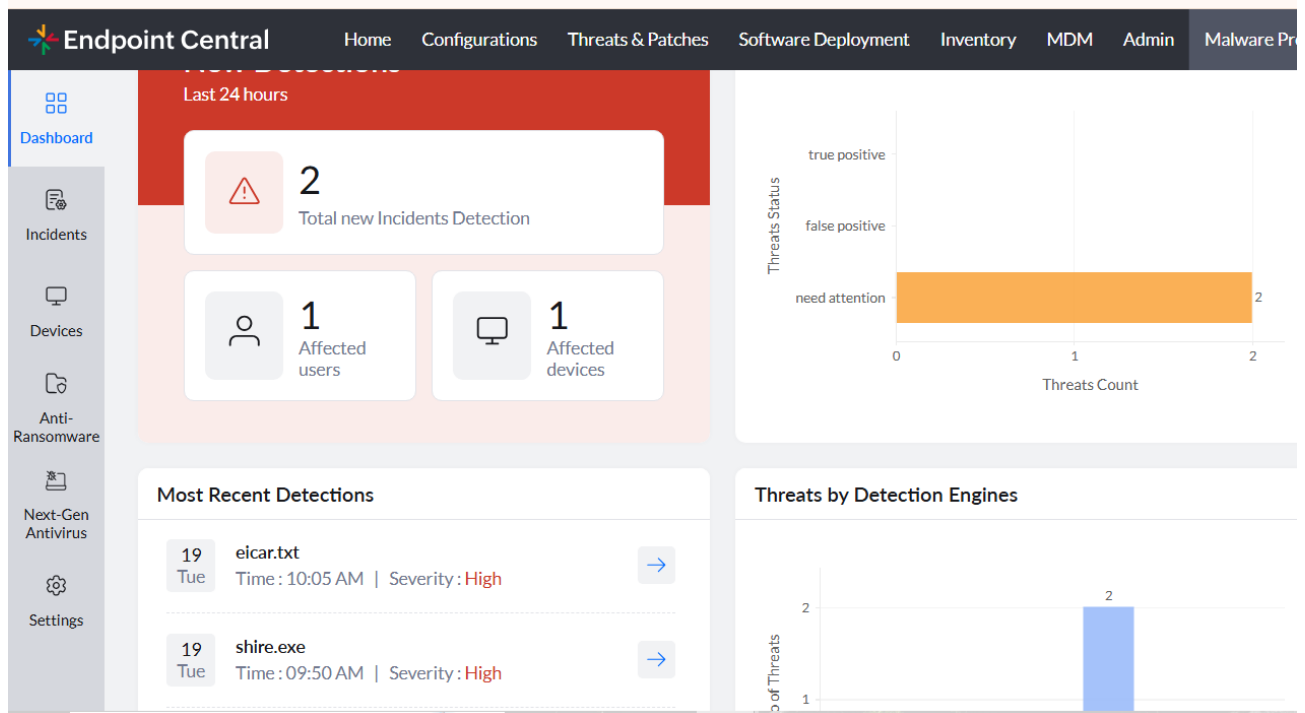
Questions & Answers

1. What threat name was detected?

The threat was flagged as **Trojan.Generic**

Which endpoint was affected?

The host device name was **Windows-The-Analyst-Endpoint**.



2. What severity level was assigned?

The severity was classified as **High**.

The screenshot shows the 'Incidents' table in Endpoint Central. The table has columns for Name, Detection Source, Detected On, Severity, Alerts, Infected Devices, Incident Status, Detection Status, and Action. The data is as follows:

Name	Detection Source	Detected On	Severity	Alerts	Infected Devices	Incident Status	Detection Status	Action
shire.exe	DeepAV Engine	May 19, 2026 09:50 AM	High	3	1	Unresolved	Open	[Resolve]
eicar.txt	DeepAV Engine	May 19, 2026 10:05 AM	High	1	1	Unresolved	Open	[Resolve]

Below the table, there is a pagination bar showing '1 - 2 of 2'.

3. Was the endpoint protected successfully?

Yes. The endpoint remained intact and uncompromised as the execution of the payload was completely arrested.

The screenshot shows the details for the 'shire.exe' incident. The top bar indicates 'Most Recent Detection: May 19, 2026 09:50 AM | Severity: High | Incident Status: Unresolved'. The 'Summary' tab is active, showing the following information:

- Signer:** Unsigned
- SHA-256:** BD0F38E9F3C7CE4570F0ED0320C9F60F0F84A9DA11C...
- Organization Information:** Company (n/a), Product (n/a), Description (n/a)
- Process Information:** File Type (EXE), Product Version (n/a), Original File Name (n/a), Internal File Name (n/a), Copyright (n/a)
- First Infected Device Information:** Device Name (The-Analyst), Time (May 19, 2026 09:50 AM)

At the bottom right, there is a 'Resolve' button and a Windows logo.

4. Was remediation successful?

Yes. The suspicious binary shire.exe was removed from the active system path and successfully moved into the encrypted quarantine area.

Endpoint Central

Home Configurations Threats & Patches Software Deployment Inventory MDM Admin Malware Protection Agent

Jump to SDP

shire.exe

Most Recent Detection: May 19, 2026 09:50 AM | Severity: High | Incident Status: Resolved

Marked as: True Positive

Summary Alerts (3) Devices (1)

Signer: Unsigned

Signed: No

Sign Algorithm: n/a

Signer Name: n/a

SHA-256: BD0F38E9F3C7CE4570F0ED0320C9F60F0F84A9DA11...

Organization Information

Company: n/a

Product: n/a

Description: n/a

Process Information

File Type: EXE

Product Version: n/a

Original File Name: n/a

Internal File Name: n/a

Copyright: n/a

First Infected Device Information

Device Name: The-Analyst

Time: May 19, 2026 09:50 AM

Incident Notes: --

Activate Windows

Go to Settings to activate Windows.

Part 5: Network Investigation

Using Sysmon network telemetry (Event ID 3), we analyzed any active connection attempts initiated by the executable.

Questions & Answers

1. Which destination IP did the malware contact?

The malware attempted to open a network connection back to **192.168.10.40**.

2. Which destination port was used?

Port **8080**.

3. Was the communication internal or external?

The network traffic stayed **internal** since the attacker and victim endpoints resided on the local testbed subnet IP range (192.168.10.1/24).

4. Why is outbound traffic monitoring important?

Monitoring outbound connections is essential to identify indicators of Compromise (IOCs) such as Command and Control (C2) callback beacons, active reverse shell payloads, unauthorized data exfiltration, or network scanning operations.

Part 6: Incident Response & Prevention

My analysis concludes with the incident containment process and broader corporate cybersecurity recommendations.

Questions & Answers

1. What immediate response actions should analysts take?

- Disconnect the target system from the active local network to isolate the endpoint.
- Terminate active suspicious system execution trees related to the identifier hashes.

- Capture dynamic memory/system diagnostic snapshots for forensically sound root cause analysis.

2. Why is Endpoint Central important in malware defense?

It acts as the local system guardrail, providing autonomous scanning engines, execution blocking filters, device isolations, and patch enforcement to protect endpoints from day one exploits.

3. How does Sysmon support incident investigations?

Sysmon delivers granular system behavioral logging (such as command execution flags, active network sockets, and file mutations) that standard Windows event tracking ignores.

4. Why is SIEM integration important?

The SIEM (Splunk) normalizes and joins multi-platform logging source into a unified query environment, enabling correlation engines to flag complex multi-step attacks.

5. How can organizations prevent similar attacks?

By maintaining software patches, locking down user administrative execution rights, implementing strict internal firewalls, and filtering incoming web traffic at proxy layers.

Bonus Threat Hunting Challenge

The following Splunk queries were developed to proactively hunt malicious payloads and active execution flags across all enrolled client devices:

1. Detect Executable Downloads

Tracks file creation processes writing binary executables to typical target user folders:

```
index=* EventCode=11 TargetFilename="*.exe"
```

The screenshot shows the Splunk Search interface. At the top, the search bar contains the query: `index=* EventCode=11 TargetFilename="*.exe"`. Below the search bar, there are tabs for "Events (12)", "Patterns", "Statistics", and "Visualization". The "Events (12)" tab is selected. The search results are displayed in a table with columns for "Time" and "Event". The first event is dated 5/19/26 at 11:37:10.120 AM. The event details show a Sysmon event where a file named "shire.exe" was created by "THE-ANALYST".

2. Detect Suspicious PowerShell Commands

Identifies command arguments containing hidden parameter structures, bypass flags, or download statements:

```
index=* (Image="*powershell.exe" OR Image="*pwsh.exe")
```

New Search Save As Create Table View Close

Index=* [Image="powershell.exe" OR Image="pwsh.exe"] Last 30 days

218 of 13,599 events matched No Event Sampling Job II Smart Mode

Events (218) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 day per column

May 19, 2026

List Format 20 Per Page Prev 1 2 3 4 5 6 7 8 ... Next

Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS			5/19/26	
a CommandLine 5			10:05:15.114 AM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}' /><EventID>13</EventID><Version>2</Version><Level>4</Level><Task>13</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-05-19T07:05:15.1198889Z' /><EventRecordID>379378</EventRecordID><Correlation><Execution ProcessID='3432' ThreadID='3676' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>The-Analyst</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='EventType'>SetValue</Data><Data Name='UtcTime'>2026-05-19 07:05:15.114</Data><Data Name='ProcessGuid'>{b6ef8e60-0391-6a0c-f61a-000000004600}</Data><Data Name='ProcessId'>15812</Data><Data Name='Image'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='ProcessName'>powershell.exe</Data></EventData></Event>

3. Detect External/Egress Communications

Tracks outbound TCP network connections exiting standard security subnets:

```
index=*EventCode=3 DestinationIp!=127.0.0.1 DestinationIp!=192.168.1.*
```

New Search Save As Create Table View Close

Index=*EventCode=3 DestinationIp!=127.0.0.1 DestinationIp!=192.168.1.* Today

✓ 437 events (5/19/26 12:00:00.000 AM to 5/19/26 1:03:00.000 PM) No Event Sampling Job II Smart Mode

Events (437) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect 1 hour per column

May 19, 2026

List Format 20 Per Page Prev 1 2 3 4 5 6 7 8 ... Next

Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS			5/19/26	
a host 1			12:57:35.561 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385f-c22a-43e0-bf4c-06f5698ffbd9}' /><EventID>3</EventID><Version>5</Version><Level>4</Level><Task>3</Task><Opcode>0</Opcode><Keywords>0x8000000000000000</Keywords><TimeCreated SystemTime='2026-05-19T09:57:00.2574976Z' /><EventRecordID>381778</EventRecordID><Correlation><Execution ProcessID='3432' ThreadID='5376' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>The-Analyst</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'></Data><Data Name='UtcTime'>2026-05-19 09:57:35.561</Data><Data Name='ProcessGuid'>{b6ef8e60-f3ff-6a0b-a801-000000004600}</Data><Data Name='ProcessId'>15812</Data><Data Name='Image'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='ProcessName'>powershell.exe</Data></EventData></Event>

4. Hunt Specific Malware Activity Across Endpoints

Counts and groups system occurrences of the specific executable payload across the environment:

```
index=* EventCode=1 Image="*shire.exe" | stats count by host, User, CommandLine
```


New Search

```
index=* EventCode=1 Image="*shire.exe" | stats count by host, User, CommandLine
```

✓ 4 events (5/19/26 12:00:00.000 AM to 5/19/26 1:05:46.000 PM) No Event Sampling ▼

Events Patterns **Statistics (4)** Visualization

20 Per Page ▼

Format

Preview ▼

host	User	CommandLine
THE-ANALYST	THE-ANALYST\The Analyst	"C:\Users\The Analyst\Downloads\lab\j\shire.exe"
THE-ANALYST	THE-ANALYST\The Analyst	"C:\Users\The Analyst\Downloads\lab\shire.exe"
THE-ANALYST	THE-ANALYST\The Analyst	"C:\Users\The Analyst\Downloads\lab\shire.exe" /silent
THE-ANALYST	THE-ANALYST\The Analyst	"C:\Users\The Analyst\Downloads\shire.exe\shire.exe"

5. Identify Blocked Malware Attempt Events

Parses the Endpoint Central index for containment events with active blocking or quarantine tags:

```
index="manageengine" sourcetype="manageengine:ec:actionlogdata"("virus" OR "malware" OR "quarantine" OR "blocked")
```

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS a host 1 a source 2 a sourcetype 1	INTERESTING FIELDS a application 1 a computerName 2 a domainName 2 # endTime 5 a endTimeISO 5 # eventId 2 # eventLogId 5 # eventTime 5 a eventTimeISO 5 a hostName 1 a index 1 # linecount 1 a module 1 a priority 1	>	5/19/26 2:40:19.730 PM	{ [-] application: ManageEngine Endpoint Central computerName: -- domainName: -- endTime: 1779188049726 endTimeISO: 2026-05-19T13:54:09.726+03:00 eventId: 482 eventLogId: 4522 eventTime: 1779188049730 eventTimeISO: 2026-05-19T13:54:09.730+03:00 hostName: -- module: Malware Protection priority: Information remarks: The ticket status of incident shire.exe modified to true positive startTime: 1779188049726 startTimeISO: 2026-05-19T13:54:09.726+03:00 timeDuration: 0 type: Information userIP: 127.0.0.1 userName: admin viewerIP: --

Indicators of Compromise (IOC) Summary

The security team has collected the following structured indicators of compromise from this lab analysis for integration into network security blocking rule-sets:

IOC Type	Value
Malware File	shire.exe
Source IP	192.168.10.40
Delivery Port	8080
Sysmon Event ID	1
Security Tool	Endpoint Central
Monitoring Tool	Splunk + Sysmon